

Stock Market Return Distribution Analyzer

Srikar Jonnalagadda
Dept. of Computer Science
and Engineering
KLH Bowrampet Campus
Hyderabad, India
Roll No. 2420030111

G Venkata Varun
Dept. of Computer Science
and Engineering
KLH Bowrampet Campus
Hyderabad, India
Roll No. 2420030187

Kurra Pranav
Dept. of Computer Science
and Engineering
KLH Bowrampet Campus
Hyderabad, India
Roll No. 2420030228

B Prathusha
Dept. of Computer Science
and Engineering
KLH Bowrampet Campus
Hyderabad, India
Project guide

Abstract - Historical stock prices reveal market direction but provide limited information about the distribution of returns and the frequency of extreme movements. This paper presents the Stock Market Return Distribution Analyzer, an interactive system for examining daily, weekly and monthly returns through transparent statistical methods. The project is justified by the need to translate established research on volatility, asymmetry and heavy tails into an accessible tool that students and beginning investors can use without advanced modelling expertise. Stocks remain the principal focus, while Bitcoin, Ethereum and the NIFTY 50 index extend comparisons across asset classes. Historical closing prices are cleaned, aligned to a common study window and converted into returns at each selected horizon. The analyzer reports the mean, median, variance, standard deviation, skewness and excess kurtosis, together with histograms, fitted Normal reference curves, extreme movement summaries and risk versus return plots. Returns beyond three standard deviations are flagged for descriptive comparison, and the largest gains and losses are displayed. Differences between cryptocurrency and exchange trading calendars are explicitly recognized, while weekly and monthly observations use period ending prices. The statistical modules are assessed using synthetic Normal and Student's t series with known distributional properties. The reported synthetic example produces daily excess kurtosis of 2.66 and an extreme return frequency of 1.04 percent for the heavy tailed series, compared with 0.09 and 0.35 percent for the Normal series. These values illustrate verification behavior rather than empirical findings about actual financial markets. The analysis also emphasizes uncertainty in monthly estimates and the sensitivity of higher moments to extreme observations. The proposed implementation combines Python, pandas, SciPy and an interactive dashboard to make every displayed statistic traceable to a standard calculation. The contribution is an integrated, explainable workflow rather than a new forecasting model. A complete market data evaluation remains necessary to answer the research questions about differences among assets and horizons. Future enhancements include quantile comparison plots, normality testing, historical and conditional value at risk, alternative distribution fitting, wider asset coverage and comparisons across market regimes. Overall, the system supports reproducible descriptive analysis while acknowledging data quality, currency differences, calendar effects and sampling uncertainty as limitations that constrain interpretation of its outputs.

Keywords - return distribution, skewness, kurtosis, volatility, extreme movements, cryptocurrency, NIFTY 50, multi-horizon analysis, data science

Stock prices are usually studied as charts, but investors and analysts make decisions about risk, and risk lives in returns rather than in prices. A price chart shows direction over time; it does not show how returns are distributed, how often large losses occur, or how one asset compares with another. Classical models often assume that returns follow a Normal distribution, yet empirical studies repeatedly find that large moves occur more often than a Normal distribution predicts [1], [2]. Describing return behavior therefore needs more than an average and a standard deviation: the asymmetry (skewness) and the tail weight (kurtosis) of the distribution must also be measured.

The project is justified by the gap between detailed research on return distributions and accessible tools for applying those findings to selected assets. Price charts alone cannot explain asymmetry, tail weight or the frequency of unusually large losses. A single workflow that calculates returns, summarizes their distributions and displays extreme observations reduces repetitive manual work and makes the analysis easier to check. Comparing stocks, cryptocurrencies and a market index at several horizons also provides a practical educational setting for understanding how risk descriptions depend on asset selection and sampling frequency. Dehouche [3] and Liu and Tsyvinski [4] motivate this attention to changes across daily, weekly and monthly horizons.

Much recent data science work on markets concentrates on forecasting with complex machine-learning models. Descriptive analysis of return distributions, however, remains the foundation of risk assessment, and it can be explained, checked and reproduced by a student or an investor. This paper therefore follows a deliberately simple and transparent route: standard statistics, standard plots and open-source tools.

This paper presents the Stock Market Return Distribution Analyzer, an interactive analytical system that converts historical prices into daily, weekly and monthly returns, measures their distribution, visualizes extreme movements and compares assets. Stock analysis is the core; cryptocurrency (Bitcoin and Ethereum) and a market index (NIFTY 50) are added to compare behavior across asset classes. The research question is: How does return behavior and distribution differ across financial asset classes and time horizons? The contributions are: (i) an integrated framework that combines return-distribution statistics, multi-period analysis and cross-asset comparison in one interactive system; (ii) a simple, explainable definition of extreme movement and a calendar-aware comparison procedure for assets that trade on different days; and (iii) a verification of the statistical modules on synthetic data with known properties, together with an evaluation plan for market data.

Massing and Ramos [2] fitted several distribution models to daily and hourly log-returns of equity indices from different countries and compared the fits using the Kolmogorov-Smirnov and Anderson-Darling statistics and the AIC and BIC criteria. A mixture of three Student's t distributions fitted best, and its components described the extreme, moderate and small returns. Welch [1] used historical U.S. stock returns to derive a simple adjustment to Z-scores for left-tail probabilities and showed that large losses occur more often than the Normal distribution predicts, with returns behaving roughly like a Student's t distribution with two to five degrees of freedom. Eom et al. [5] measured tail fatness and found that diversification does

not remove it: in Korean data the positive tail thins while the negative tail grows fatter as stocks are added, which helps to explain losses in crashes. Eom, Eom and Park [6] showed that the tail properties of individual Korean stocks differ across the cross-section and must be considered when studying left-tail momentum.

Wątorrek et al. [7] analyzed 2017-2020 return distributions of currencies, cryptocurrencies and contracts for difference on stocks, indices and commodities, modelling the tails with power-law, stretched exponential and q-Gaussian functions. They found that the inverse-cubic power law is still a fair reference at short horizons and that the earlier trend towards faster convergence to normality is no longer seen. Sharma [8] built empirical return distributions of the Nifty 50 for horizons from one day to twelve years; one-day returns were mildly positively skewed with 56 percent of days positive, while one-year returns showed a 74 percent chance of gain. Studies of Bitcoin add that behavior depends on the sampling frequency [3], [4].

Mamilla et al. [9] applied ARCH and GARCH models to nine NSE indices before, during and after COVID-19. The Nifty Realty index was the most volatile, and the Bank, Metal and IT indices gave higher returns during COVID-19 than before. Mata et al. [10] fitted a multivariate Normal Inverse Gaussian distribution to BRIC stock market returns, found evidence of integration among these markets, and reported that Value at Risk (VaR) performed better with this distribution than with multivariate GARCH models.

Liu et al. [11] introduced a quantile-based robust kurtosis measure and tested it on monthly returns of 39 country indices from 1996 to 2019. Higher robust kurtosis predicted lower future returns, mainly in developed markets, and robust skewness carried a positive premium, mainly in emerging markets. Kan and Lassance [12] showed that fat tails worsen the out-of-sample performance of sample mean-variance and global minimum-variance portfolios and that accounting for fat tails improves it.

Table I summarizes the reviewed studies. Three observations follow. First, the literature is rich in tail models and in forecasting or pricing methods, but each study typically centers on one market, one asset type or one modelling method. Second, the mean and standard deviation alone miss the asymmetry and fat tails that these studies emphasize [1], [5]. Third, the findings are published as research methods, not as simple reusable analyzers that a student or an investor can run on chosen assets and horizons. The gap addressed here is therefore not a new statistical model but the integration of return calculation, distribution statistics, visualization of extremes, multi-period analysis and cross-asset comparison into one interactive and explainable system.

Table I. Reviewed studies and their limitations for this work

Study	Focus	Limitation
Massing and Ramos [2]	Index return distribution models	Model fitting; no interactive tool
Wątorrek et al. [7]	Multiple asset types and tail models	No daily, weekly and monthly stock comparison
Mata et al. [10]	BRIC dependence and value at risk	No single stock visuals
Eom et al. [5]	Tail fatness under diversification	Northeast Asian portfolio evidence
Eom, Eom and Park [6]	Korean left-tail momentum	Single market and momentum focus

Study	Focus	Limitation
Mamilla et al. [9]	GARCH for nine NSE indices	Volatility forecasting focus
Welch [1]	Adjustment to left-tail probabilities	Left tail; no cross-stock tool
Liu et al. [11]	Robust kurtosis of country indices	Research method; working paper
Kan and Lassance [12]	Fat tails and portfolio performance	Portfolio weights, not asset distributions
Sharma [8]	Nifty 50 at multiple horizons	Single index; preprint

The project addresses four connected problems: raw prices hide return behavior; risk requires more than an average; extreme movements can be missed without distribution plots; and assets and horizons are difficult to compare without a consistent method. Variance, standard deviation, skewness and kurtosis describe complementary aspects of the distribution, but repeated manual calculation is slow and prone to error.

A simple analysis system is therefore needed that turns price history into daily, weekly or monthly returns and organizes them for statistical analysis and comparison.

The research question is: How does return behavior and distribution differ across financial asset classes and time horizons? It is divided into three sub-questions. RQ1: How do the mean, volatility, skewness and kurtosis differ between stocks, cryptocurrency and a market index? RQ2: How do these measures change from daily to weekly to monthly returns, and do the returns move closer to a Normal shape? RQ3: How often, and how large, are extreme movements in each asset class? The objectives are to compute daily, weekly and monthly returns for stocks, cryptocurrency and NIFTY 50; organize the data for statistical analysis; measure the mean, median, variance, standard deviation, skewness and kurtosis; visualize distributions and extreme movements; compare asset classes and horizons; and relate average return to volatility.

Daily closing prices are collected from Yahoo Finance [13]. Stocks form the core of the study, and Table II lists the asset classes. The stocks are three to four NSE-listed large-cap companies from different sectors. Bitcoin and Ethereum represent cryptocurrency, and NIFTY 50 represents a market index. A common window of about eight years (for example, January 2018 to the latest complete month) is used so that every asset, including Ethereum, has data for the whole period; availability of each series will be confirmed before the final run. Eight years give about 2000 daily, 400 weekly and 96 monthly observations.

Table II. Asset classes used in the study

Asset class	Examples	Calendar / unit
Stocks (core)	Three to four NSE large-cap stocks from different sectors	Exchange trading days / INR
Cryptocurrency	Bitcoin, Ethereum	Every day / USD
Market index	NIFTY 50	Exchange trading days / index points

Duplicate dates are removed, missing prices are dropped rather than filled, and all series are trimmed to the common window. Cryptocurrency trades every day whereas stocks and the index trade only on weekdays, so daily returns of the two groups are not measured on identical calendars [3]. Daily statistics are therefore computed on each asset's own calendar and interpreted with this caveat, while weekly and

monthly returns use the last available closing price of each week or month, which provides consistent period labels while differences in closing times and nontrading days remain. All analysis uses percentage returns, so the quote unit does not affect the results; the exchange-rate effect of comparing USD-quoted and INR-quoted assets is listed as a limitation.

For closing price P at the end of period t , the simple return is calculated as a fraction using (1). Multiplying this value by 100 gives the percentage return shown in the dashboard.

$$R(t) = [P(t) - P(t-1)] / P(t-1) \quad (1)$$

Weekly and monthly returns use the last available closing price of each period; daily simple returns are compounded, not added, when constructing an equivalent multi-period return. Simple returns are easy to interpret and approximate log returns only when movements are small [2], [3].

For n observations, the sample mean, sample standard deviation and standardized return are calculated as follows:

$$\text{mean} = [R(1) + R(2) + \dots + R(n)] / n \quad (2)$$

$$s = \sqrt{\sum[(R(t) - \text{mean})^2] / (n - 1)} \quad (3)$$

$$z(t) = [R(t) - \text{mean}] / s \quad (4)$$

The sample variance is s squared. The median is the middle ordered observation, or the average of the two middle observations. For skewness and excess kurtosis, define the central moment $m(k)$ as the average of $(R - \text{mean})$ raised to power k . The uncorrected moment coefficients are:

$$g = m(3) / m(2)^{3/2} \quad (5)$$

$$K = m(4) / m(2)^2 - 3 \quad (6)$$

The implementation uses the corresponding bias corrected estimators provided by pandas and SciPy [14], [15], rather than interpreting these uncorrected coefficients as corrected sample statistics. Excess kurtosis is zero for a Normal population; positive estimates indicate greater fourth moment tail weight relative to this reference. Standard deviation measures volatility. Constant series require special handling because standardized returns and the return to risk ratio are undefined when s is zero.

Table III. Statistics and their interpretation

Statistic	Interpretation
Mean	Average return per period
Median	Middle return; less affected by extremes
Variance	Squared spread around the mean
Standard deviation	Spread of returns; volatility measure
Skewness	Asymmetry; negative values suggest a longer left tail
Excess kurtosis	Fourth moment tail weight relative to a Normal population

A return is called an extreme movement when its standardized value satisfies $|z| > 3$, that is, when it lies more than three standard deviations from the mean. For a Normal distribution about 0.3 percent of returns fall outside this band, so the observed share of extreme movements can be compared directly with this benchmark. The analyzer also lists the five best and five worst returns of each asset and horizon, and overlays a Normal curve with the same mean and standard deviation on every histogram so that fat tails

are visible.

Each asset and horizon is placed on a plot of mean return against standard deviation. A simple return-to-risk ratio

$$RR = \text{mean} / s \quad (7)$$

summarizes the return earned per unit of volatility. It is not annualized and uses no risk-free rate, so it is used only to compare assets within the same horizon.

Fig. 1 shows the five stages of the analyzer. The pipeline is implemented in Python. Data handling uses pandas [14], the statistics use SciPy [15], and plots are drawn with Matplotlib or Plotly. A Streamlit dashboard lets the user choose the asset or assets, the return horizon (daily, weekly or monthly) and the time period, and then shows the statistics table, the histogram with the Normal curve, the extreme-movement summary and the risk versus return plot. Every output is produced by a standard formula described above, so each number on the dashboard can be traced and explained.

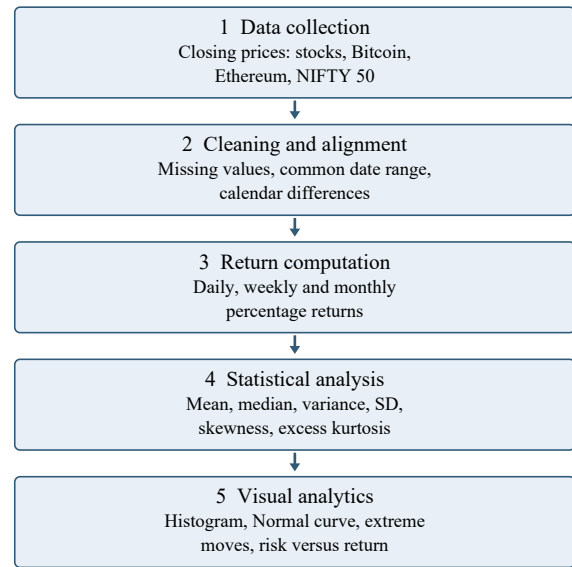


Fig. 1. Processing pipeline of the Stock Market Return Distribution Analyzer.

Optional extensions are kept separate from the core system and are implemented only if time permits: a Q-Q plot for visual checking of normality, the Jarque-Bera normality test, which uses skewness and kurtosis, and historical VaR, defined as the 5th-percentile return. Student's t fitting, conditional VaR and comparison of sub-periods such as the COVID-19 period [9] are left as future work. Forecasting models, sentiment analysis and portfolio optimization are outside the scope of this work.

Before market data are analyzed, the statistical modules are verified on synthetic returns whose properties are known. Two series of 2016 daily returns (about eight years of 252 trading days) were generated with a fixed random seed and a daily standard deviation of 1 percent: one from a Normal distribution and one from a fat-tailed Student's t distribution with five degrees of freedom, rescaled to the same standard deviation. Weekly and monthly synthetic series were formed by summing blocks of 5 and 21 consecutive daily values to test additive aggregation. This verification is distinct from the compounded simple return calculation used for market prices. These data are artificial and are used only to test the implementation; they are not market results.

Table IV. Synthetic verification of the statistics

Data	Horizon	n	SD (%)	Skew	Ex. kurt.	>3 SD (%)
Normal	Daily	2016	1.00	0.00	0.09	0.35
Normal	Weekly	403	2.23	0.19	0.13	0.25
Normal	Monthly	96	4.53	-0.09	-0.03	0.00
Fat-tailed	Daily	2016	0.99	-0.07	2.66	1.04
Fat-tailed	Weekly	403	2.13	-0.14	0.27	0.25
Fat-tailed	Monthly	96	4.37	-0.15	-0.55	0.00

SD: standard deviation; Ex. kurt.: excess kurtosis; >3 SD: percentage beyond three standard deviations.

Table IV reports the synthetic verification results. These examples are consistent with the expected qualitative contrast between the two generating distributions. For the Normal series, skewness and excess kurtosis are close to zero and the daily share of extreme movements is 0.35 percent, near the 0.27 percent expected. For the fat-tailed series, the daily excess kurtosis is 2.66 and 1.04 percent of returns are extreme, about four times the Normal benchmark. Fig. 2 shows the corresponding analyzer output. When independent returns are summed over longer periods, excess kurtosis falls roughly in proportion to one divided by the number of periods summed, and the observed excess kurtosis is smaller at the weekly and monthly horizons (0.27 and -0.55). These estimates alone do not establish normality. This gives a useful reference for the market study: excess kurtosis that persists at weekly or monthly horizons in real data may motivate investigation of volatility clustering or jumps, although finite sample variation must also be considered.

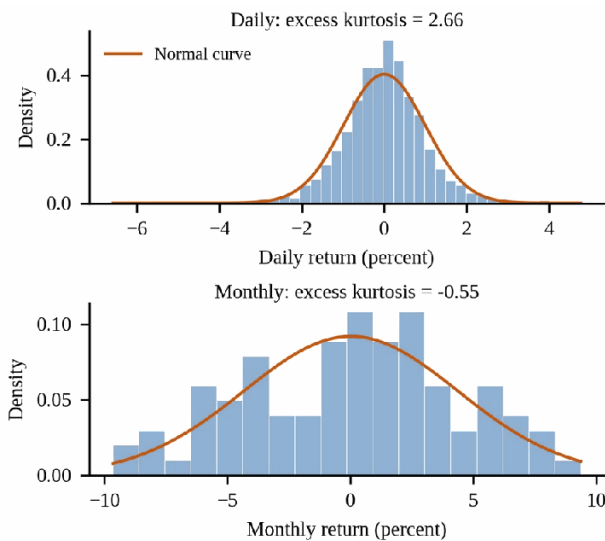


Fig. 2. Analyzer output on the synthetic fat-tailed series: daily and monthly histograms with a Normal curve.

The verification also shows how much sampling noise to expect. Over 2000 repeated Normal samples of 96 monthly observations, the middle 90 percent of the excess kurtosis values ranged from -0.65 to 0.89 and that of skewness from -0.41 to 0.41, although the true values are zero. For the fat-tailed daily series the 90 percent range of excess kurtosis was 1.73 to 11.33. Monthly statistics from about 100 observations, and kurtosis in general, must therefore be reported with this uncertainty in mind and not read as exact values.

The analyzer will be run on the assets in Table II, and the results will be read against the three sub-questions. For RQ1,

the statistics tables of the three asset classes are compared at each horizon. Cryptocurrency is expected to show larger dispersion than stocks and the index [3], [4], and the index is expected to be less volatile than a typical single stock because it averages many stocks; both are hypotheses to be tested and not assumptions. For RQ2, the change of skewness, kurtosis and standard deviation from daily to monthly returns is compared across asset classes. No single outcome is presumed, because the literature reports mixed evidence: excess kurtosis of Bitcoin fell with longer horizons while the opposite tail behavior was reported for S&P 500 [3], and convergence to normality is not always observed [7]. For RQ3, the share of returns beyond three standard deviations and the worst returns are compared with the 0.3 percent Normal benchmark and with the synthetic reference reported above.

The study is descriptive and makes no forecast. Its findings apply to the chosen period and assets, and market regimes such as the COVID-19 period [9] may influence the statistics. Daily comparison between cryptocurrency and stocks is affected by the different trading calendars, and comparison of USD-quoted and INR-quoted assets ignores exchange-rate movements. Monthly samples are small, and the standard deviation and kurtosis can be dominated by a few extreme observations when tails are heavy [3]. Data quality depends on the public data source. These limits will be stated with every reported result.

For students and beginner investors, the analyzer turns a price history into a small set of interpretable numbers and plots. For researchers, it offers a reproducible baseline analysis of return distributions across asset classes and horizons before any advanced model is applied.

Conclusion and future enhancements

This paper presented the Stock Market Return Distribution Analyzer, which integrates return-distribution statistics, multi-period analysis and cross-asset comparison in one interactive and explainable system. Stocks remain the core, and cryptocurrency and NIFTY 50 extend the analysis to answer the question of how return behavior and distribution differ across asset classes and time horizons. The design, the definitions and the calendar-aware comparison procedure were described, and the statistical modules were verified on synthetic data. The next step is the full run on market data, whose results will answer RQ1 to RQ3. Future enhancements include the optional Q-Q plot, Jarque-Bera test and historical VaR, Student's *t* fitting, conditional VaR, sub-period comparison and a wider set of assets. Further improvements should document the exact data extraction date, provide a common currency comparison option and report uncertainty intervals for distributional statistics. These enhancements would strengthen reproducibility and help users distinguish sampling variation from persistent market behavior.

The authors thank the Department of Computer Science and Engineering, KLH Bowrampet Campus, for the facilities and support provided for the Data Science (24AD3107) project.

References

- [1] I. Welch, "A heuristic for fat-tailed stock market returns," *Financial Analysts J.*, vol. 80, no. 4, pp. 18-26, 2024.
- [2] T. Massing and A. Ramos, "Student's t mixture models for stock indices: A comparative study," *Physica A*, vol. 580, Art. no. 126143, 2021.
- [3] N. Dehouche, "Scale matters: The daily, weekly and monthly volatility and predictability of Bitcoin, Gold, and the S&P 500," 2021, arXiv:2103.00395. [Online]. Available: <https://arxiv.org/abs/2103.00395>
- [4] Y. Liu and A. Tsyvinski, "Risks and returns of cryptocurrency," *Nat. Bureau Econ. Res.*, Cambridge, MA, USA, Working Paper 24877, 2018. [Online]. Available: https://www.nber.org/system/files/working_papers/w24877/w24877.pdf
- [5] C. Eom, T. Kaizoji, G. Livan, and E. Scalas, "Limitations of portfolio diversification through fat tails of the return distributions: Some empirical evidence," *N. Am. J. Econ. Finance*, vol. 56, 2021.
- [6] C. Eom, Y. Eom, and J. W. Park, "Left-tail momentum and tail properties of return distributions: A case of Korea," *Int. Rev. Financial Anal.*, vol. 87, Art. no. 102570, May 2023.
- [7] M. Wątopek, J. Kwapien, and S. Drożdż, "Financial return distributions: Past, present, and COVID-19," *Entropy*, vol. 23, no. 7, Art. no. 884, 2021.
- [8] C. Sharma, "Multi-scale analysis of Nifty 50: Return characteristics, valuation dynamics and market complexity (1990-2024)," 2025, arXiv:2509.00697. [Online]. Available: <https://arxiv.org/abs/2509.00697>
- [9] R. Mamilla, C. Kathiravan, A. Salamzadeh, L.-P. Dana, and M. Elheddad, "COVID-19 pandemic and indices volatility: Evidence from GARCH models," *J. Risk Financial Manag.*, vol. 16, no. 10, Art. no. 447, Oct. 2023.
- [10] L. Mata Mata, J. A. Núñez Mora, and R. Serrano Bautista, "Multivariate distribution in the stock markets of Brazil, Russia, India, and China," *SAGE Open*, vol. 11, no. 2, Art. no. 21582440211009509, Apr. 2021.
- [11] R. Liu, A. Maynard, and I. Tsiakas, "Robust conditional kurtosis and the cross-section of international stock returns," unpublished working paper, Jun. 2024.
- [12] R. Kan and N. Lassance, "Optimal portfolio choice with fat tails and parameter uncertainty," *J. Financial Quant. Anal.*, 2025.
- [13] Yahoo Finance, "Historical price data for Bitcoin (BTC-USD), Ethereum (ETH-USD) and NIFTY 50 (^NSEI)." [Online]. Available: <https://finance.yahoo.com>
- [14] W. McKinney, "Data structures for statistical computing in Python," in *Proc. 9th Python in Science Conf.*, 2010, pp. 56-61.
- [15] P. Virtanen et al., "SciPy 1.0: Fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261-272, 2020.